Kinetic X Studio

AI Humanoid Animator
Turn real movement into fluid 3D animation

Hacktech by Caltech 2026

Meet the team

Kelly Lwin

Background:

- Computer Science Major, Data Science Minor - Cal Poly Pomona
- AI/ML - Break Through Tech Program @Cornell Tech
- Backend - CPP Bronco Bond Student Social Networking App
- Frontend - CPP IT Department: handles university website dev

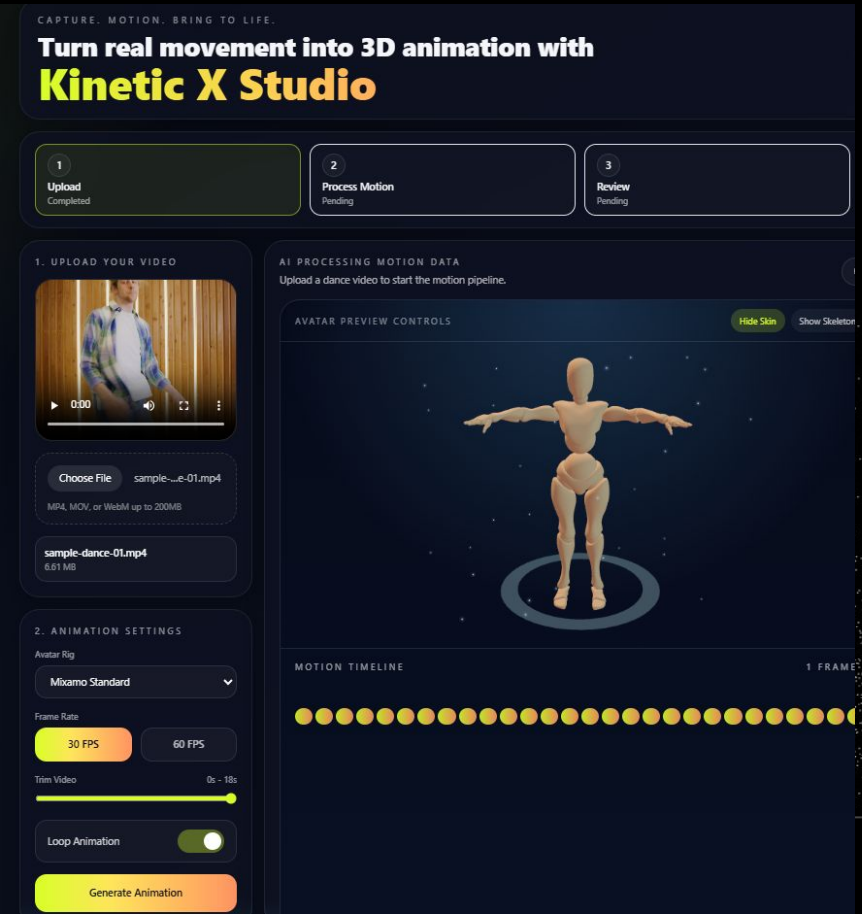
Others:

- Learning game dev with Unity
- Exp. in Blender - 3D modeling
- Building side projects

Inspiration

- Game development and Blender modeling are my hobbies
- Character animation takes significant time
- Mixamo presets are useful but limited for custom motions
- Manual keyframing requires strong animation skills
- Needed a faster way to convert real human motion into reusable animations
- Built Kinetic X Studio to solve this problem and help other creators

What it does



1. Users upload short movement videos through a web app (featuring a person)
2. AI analyzes motion and extracts body pose data
3. System converts movement into humanoid animation sequences
4. Motion is retargeted onto a standardized humanoid rig
5. Exports reusable FBX animation files Compatible with Blender, Unity, Unreal, etc.
6. Reduces manual keyframing and expensive mocap dependency

How we built it:

ChatGPT | Codex

MediaPipe (google AI)

core motion extraction engine that converts dance video into pose landmarks

Blender (python API)

runs server-side in headless mode to retarget AI-extracted dance motion onto a humanoid rig and export a reusable FBX animation file

K2 Think V2 LLM

LLM-based reasoning layer - analyzes pose data, reasons about motion quality, improves rig mapping decisions, and helps clean the animation pipeline before FBX export

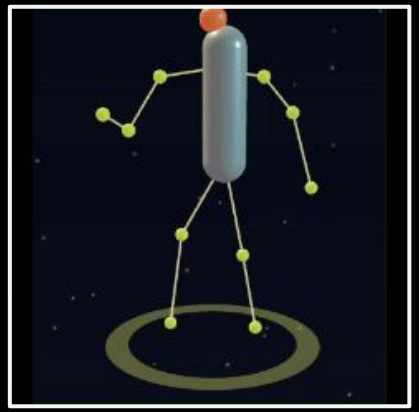
React Three Fiber

renders the live 3D humanoid avatar in the browser and applies the generated motion so users can preview the animation before downloading the FBX file

Demo Time

Challenges we ran into:

- FBX export failed from missing mesh and animation issues
- Avatar bones broke, causing merged limbs and unnatural movement
- MediaPipe failed, forcing synthetic fallback instead of real pose tracking
- K2 Think V2 API failed, breaking reasoning and motion cleanup
- Solo hackathon time limits required fast building with Codex and ChatGPT

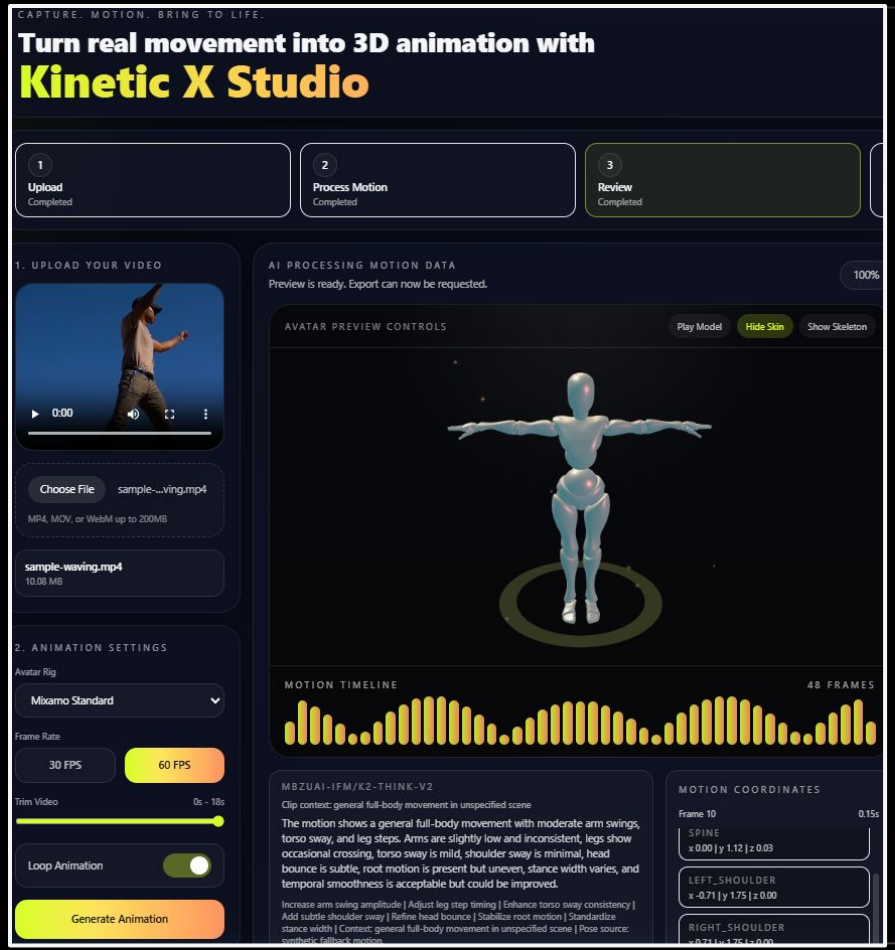


Accomplishments we are proud of

- Built a reliable MVP that maps sample videos to matching Mixamo animations on a 3D avatar
- Integrated K2 Think V2 to analyze motion context and generate clear motion labels
- Delivered full workflow from video upload to animation preview and Blender-based FBX export

What we learned

- Keeping scope small is important
- FBX export is harder than it looks
- Retargeting motion is the hardest part
- Real pose tracking is better than fallback motion
- Small backend bugs can break the whole pipeline



What's next for Kinetic X Studio

- Cloud storage and animation showcase gallery for managing saved FBX exports
- FBX sharing, collaboration, and reusable animation library support
- More avatar options, better environments, and expanded rig support for Unity, VRM, and humanoid pipelines
- Improved motion accuracy through stronger pose estimation, retargeting, and reasoning models
- More polished UX with better interactions, music, and creator-focused workflow improvements

Thank you

Project GitHub:

https://github.com/phyulwin/humanoid-motion-synthesis-fbx-generator/